# IC-37 CAPACITORS IN CIRCUITS

*3/05/2023* ***FOR IN-CLASS AND TAKE-HOME LAB KIT***

## 37-1 OBJECTIVES

The purpose of this lab is to determine how a capacitor behaves in an R-C circuit by measuring the time for charging and discharging. This will also be done for two capacitors in Series and in Parallel.

## 37-2 EQUIPMENT



AC/DC Electronics Lab Board

Capacitors

Resistors

Wire Leads

D-cell Battery

Stopwatch / Cellphone

Voltage Sensor / Digital Multimeter (DMM).

Capstone Software.



## 37-3 THEORY

An RC circuit is one in which we have a resistor in series with a capacitor (Figure 1). In this figure, the battery is not connected to the circuit, and there is no charging of the capacitor. Assume that the capacitor is initially completely un-charged (which can be done by momentarily connecting both ends of the capacitor with a piece of wire). The switch is then thrown to position 'a' at time $t$ = 0 s (figure 2). This starts charging the capacitor, and the rate of charging is given by:

$$q(t) = Q\left(1 - e^{\frac{-t}{RC}}\right) \qquad (1)$$

Where $q(t)$ is the charge at time $t$ after charging starts, $R$ and $C$ are the values of the resistance and capacitance. $Q$ is the maximum charge that can be stored on the capacitor, and is given by

$$Q = C\,\Delta V \qquad (2)$$

Where $C$ is the Capacitance of the capacitor, and $\Delta V$ is the applied potential across the capacitor (= $\varepsilon$ volts). The term $RC$ is called the time constant ( $\tau$ ) of the $RC$ circuit, and is the time taken for the capacitor to charge to 63.2% of its maximum possible value.

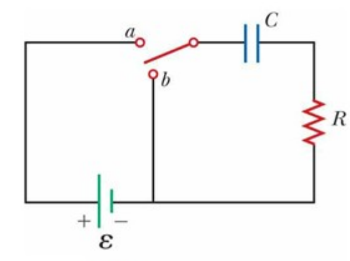



Figure 1

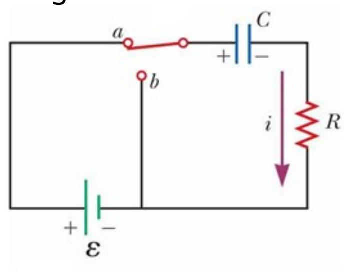



Figure 2

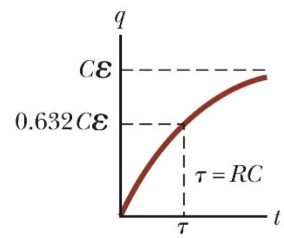



Figure 3

The capacitor charges with time as shown in Figure 3. Even though equation (1) and this curve indicate that the capacitor will never be fully charged, for most practical purposes we consider that the capacitor is fully charges after a time that is more than 5 time-constants have passed.

To discharge the capacitor through the resistor, we move the switch to position ‘b’, as in Figure 4. The capacitor discharges according to equation (3). The charge decreases exponentially as shown in Figure 5.

$$q(t) = Q e^{\frac{-t}{RC}} \qquad (3)$$

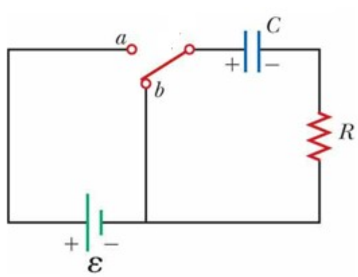



Figure 4

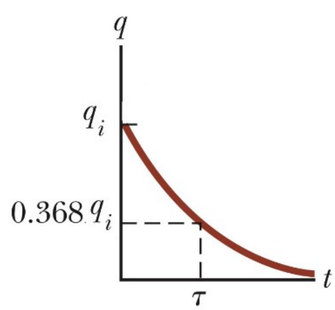



Figure 5

In this experiment we will determine the time that it takes for the capacitor to charge and discharge to some value. Instead of measuring the charge, we will measure the voltage across the capacitor, which is an indicator of how much the capacitor has charged, as a function of the maximum charge. So while charging, when the potential difference across the capacitor reaches the battery potential, the potential drop across the resistor is zero and there is no current in the circuit. This means that the capacitor is fully charged.

## 37-4 PROCEDURE

1. Connect the circuit as shown in Figure 6, using a 100 **kΩ** resistor and a 100 **μF** capacitor (or close to these values). Note that the wire sticking out of **SW2** is not connected on the other side. This will be used later to discharge the capacitor.

2. Connect the Voltage Sensor set to measure Voltage and connect it to Capstone. In case voltage sensor is not available, use the Multimeter in Voltage mode.

3. The initial few steps are to familiarize you with how the capacitor behaves. Start with no voltage on the capacitor and the switch off. If there is remaining voltage on the capacitor, use a piece of wire to “short” the two leads together, draining any remaining charge. (Touch the ends of a wire to points **C5** and **C7** as shown in Figure 6 to discharge the capacitor.)

4. Now close the switch by pushing and holding the button down. Observe how the voltage across the capacitor changes with time. The voltage should rise as shown by ‘**q**’ in Fig 3.

5. If you now open the switch by releasing the button, the capacitor should remain at its present voltage with a very slow drop over time. This indicates that the charge you placed on the capacitor has no way to move back to neutralize the excess charges on the two plates. In practice, there is some leakage of current through the Voltage Sensor (or Multimeter), and the voltage will drop slowly across the capacitor.

6. Connect a wire between the points marked as **C5** and **C8** in the circuit, allowing the charge to drain back through the resistor. This can be done by connecting the wire sticking out of **SW2** into **C5**. Observe how the voltage across the capacitor changes with time. The voltage should fall off as shown by ‘**q**’ in Fig 5.

7. Repeat steps 3-5 until you have a good feeling for the process of charging and discharging of a capacitor through a resistance.

8. With the switch in ‘open’ position (i.e. not pressed), get the battery voltage by attaching the voltmeter prongs to the ends of the battery. This is the Battery EMF, or **Vmax**.

9. Calculate the value of **0.632***__Vmax__ and call this $\mathbf{V_1}$. Calculate **0.368*Vmax** and call this $\mathbf{V_2}$.

10. Now attach the voltage sensor across the capacitor. Press ‘Record’ in Capstone to start measuring the voltage. Press the switch to start charging the capacitor and keep charging for a long time and note the maximum voltage that is reached across the capacitor. This should be the same as the battery EMF. If not, there is a leakage current through the voltmeter (since its resistance is not infinite). Use the measured maximum voltage as **Vmax**.

    If using the Multimeter, start the stopwatch and press the switch simultaneously, and note the time taken to reach $\mathbf{V_1}$**, i.e. 0.632***__Vmax__. Keep charging till the voltage changes no more (reaches **Vmax)**.

11. Taking the Data:

***Taking the data by using voltage Sensor and Capstone:***

a) With the circuit connected as in Fig. 6, discharge the capacitor by momentarily connecting **C5** and **C7** with a piece of wire.

b) Start Record in Capstone and press the switch to charge the capacitor. Fig 6.

c) Keep charging the capacitor till it is ‘fully charged’ and reaches Vmax (about 5*$\mathbf{t_C}$).

d) Simultaneously release switch and connect the wire from **SW2** to **C5**. Fig 7.

e) Stop recording when the voltage falls to about one tenth of **Vmax**.

f) Use the Coordinates Tool in Capstone to determine the time taken to go from V=0 to V= $\mathbf{V_1}$. This is the charging time $\mathbf{t_C}$.

g) Determine the time taken to go from **Vmax** to $\mathbf{V_2}$. This is the Discharging Time $\mathbf{t_D}$.

h) Note the values of $\mathbf{t_C}$ and $\mathbf{t_D}$ in Table 1.

***Taking the data by using Multimeter and Stopwatch:***

a) With the circuit connected as in Fig. 6, discharge the capacitor by momentarily connecting **C5** and **C7** by a wire.

b) Start the stopwatch and press the switch simultaneously, and note the time taken to reach $\mathbf{V_1}$**, i.e. 0.632***Vmax**. This is the charging time $\mathbf{t_C}$.

c) Turn off stopwatch. Keep charging the capacitor till it is ‘fully charged’ and reaches Vmax (about 5*$\mathbf{t_C}$).

d) Simultaneously release switch, connect the wire from **SW2** to **C5** and start the stopwatch. Note the time it takes to reach V2 i.e. 0.368 Vmax. Fig 7. This is the Discharging Time $\mathbf{t_D}$.

e) Stop recording.

f) Note the values of $\mathbf{t_C}$ and $\mathbf{t_D}$ in Table 1.

12. Repeat the processes in 11) five times, and record in Table 1.

If the time to charge and/or discharge are less than about 5 seconds, or more than a few minutes, make sure you are using the correct procedure, or change the resistor or capacitor to a different value. For the next part, pick a second capacitor that is within about half to two times the first one. It should not be too much smaller or larger.

13. Add the second capacitor in series with the first one, to make the circuit as shown in figure 8. Repeat the measurement of charging and discharging time (i.e. step 11). Save the Capstone Screenshots and record the times in Table 2.

14. Now re-arrange the circuit to make it as shown in figure 9. You now have the two capacitors in parallel. Repeat the measurement of charging and discharging time (i.e. step 11). Save the Capstone screenshots and record the times in Table 2.

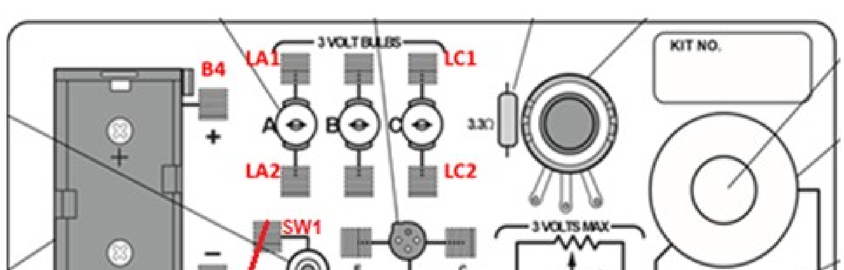
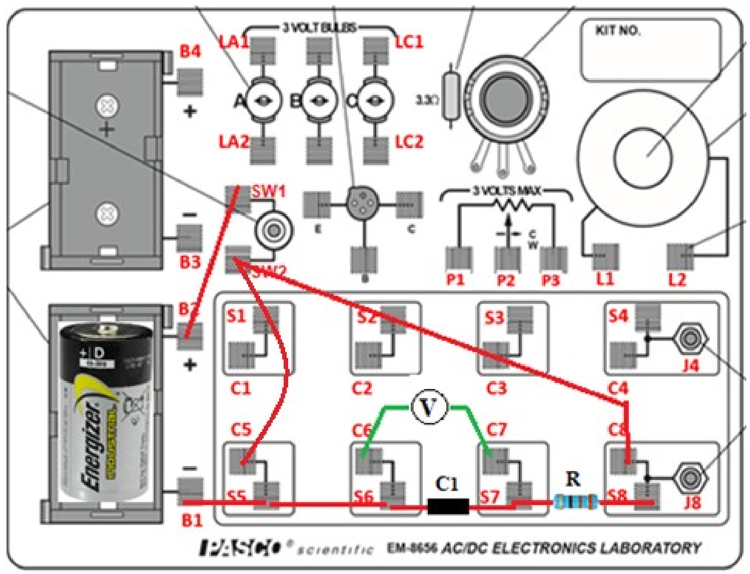


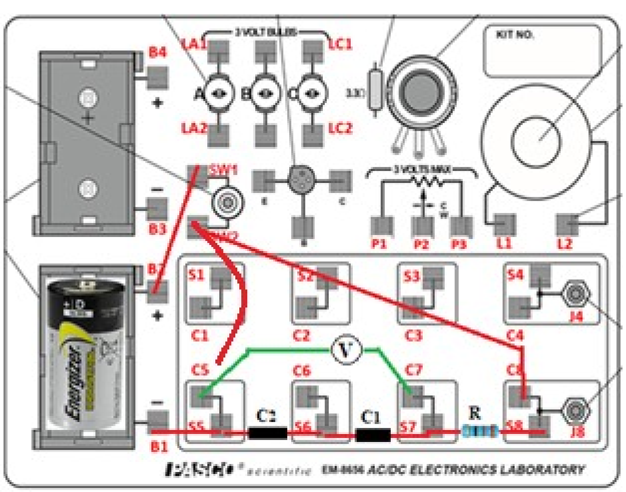
Figure 6: Single Capacitor in R-C Circuit.

Charging.

Figure 7: Single Capacitor in R-C Circuit.

Dis-charging



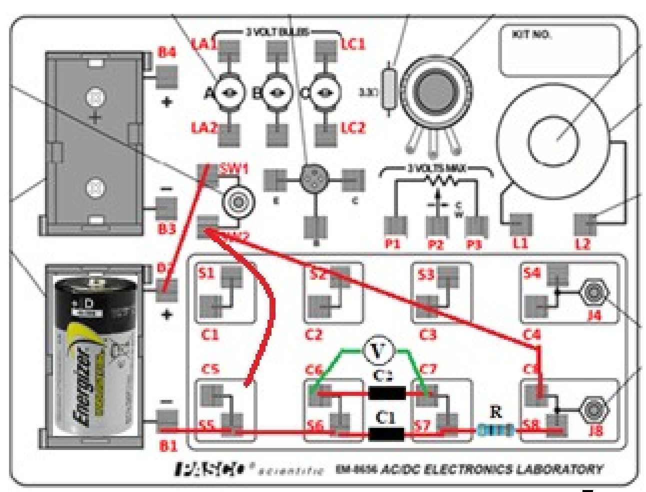



Figure 8: Capacitors in Series - charging

Figure 9: Capacitors in Parallel – charging

## 37-5 PRECAUTIONS

1. When using Capstone, you DO NOT have to press record and the switch at the same time. Press Record first, then press the switch.
2. When using the Multimeter, you DO HAVE to press the switch and start the stopwatch at the same time.
3. When discharging the capacitor, release the switch and connect the wire between switch and C5 at the same time.
4. Select the resistor and capacitors such the time period is not less than about ten seconds and not more than a few minutes.
5. When doing the series and parallel cases, the values of the two capacitors should be comparable to each other, i.e. one of them should be within 2 times the other in capacitance.

## 37-6 DATA: CAPACITORS IN CIRCUITS

Resistance: _____ Capacitance: _____ Time Constant $\tau$ = RC = _____

Battery potential: _____

***Table 1: Time for charging and discharging a capacitor***

| Trial No. | **Vmax** <br> **V** | **$V_1$ =0.632* Vmax** <br> **V** | **$V_2$ = 0.368 Vmax** <br> **V** | Measured Charging time $\mathbf{t_C}$ <br> *$0 – V_1$* <br> **s** | Measured Dischargin g time $\mathbf{t_D}$ <br> *$Vmax – V_2$* <br> **s** | Percent error in time of charging | Percent error in time of dischargin g |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

**Two Capacitors in Series and Parallel:**

Resistance: _____ Capacitor 1: _____ Capacitor 2: _____

***Table 2: Time for charging and discharging two capacitors in series and parallel.***

| Type of Circuit: Cap.s in: | Equiv. Cap. **Ceq** **μF** | Calc. time for $t_C$ and $t_D$ $=RC_{eq}$ **s** | **Vmax** **V** | $V_1$ **=0.632* Vmax** **V** | $V_2$ **= 0.368 Vmax** **V** | Meas. Charging time $t_C$ **s** | Meas. Dis-charging time $t_D$ **s** | Percent error in time of charging | Percent error in time of dis-charging |
|---|---|---|---|---|---|---|---|---|---|
| Series | | | | | | | | | |
| Parallel | | | | | | | | | |

### 37-7 REPORT:

Upload the following in the DataSet for this Lab.

| | | ***Take-Home Lab-Kit or In-Class if you are using Capstone.*** | Points |
|---|---|---|---|
| 1. | 1 | Show your circuit setup | 5 |
| 2. | 2 | Completely filled up "Report Form". Make sure to include units. | 20 |
| 3. | 3 | At least one Photograph of the circuit, showing voltage sensor in place for measurement | 5 |
| 4. | 4 | At least two Capstone screenshots, one for charging and one for discharging the capacitor (select the "best" graphs and show the positions of tc and td) | 2*5 = 10 |
| 5. | 5 | Sample calculations, one for charging time and one for discharging time | 5 |
| 6. | 6 | Sources of Error in this experiment. | 5 |
| 7. | 7 | Discussion of Results | 10 |
| | | Total | 60 |

| | ***In-Class if you are using the Multimeter and Stopwatch*** | Points |
|---|---|---|
| 1 | Show your circuit setup | 5 |
| 2 | Completely filled up "Report Forms". Make sure to include units. | 20 |
| 3 | At least one Photograph of the circuit, showing multimeter in place for measurement | 5 |
| 4 | Sample calculations, one for Charging time and one for discharging time | 5 |
| 5 | Sources of Error in this experiment. | 5 |
| 6 | Discussion of Results | 10 |
| | Total | 50 |

### 37-8 ADDITIONAL INFORMATION

RC Circuit

https://www.khanacademy.org/science/electrical-engineering/ee-circuit-analysis-topic/ee-natural-and-forced-response/v/ee-rc-natural-response-intuition

### 37-9 POINTS TO THINK ABOUT

1. Why are the times to charge and discharge the capacitor the same?
2. Why is the time to charge (and discharge) more for the parallel combination than for the series combination?
3. Why should the two capacitors be of comparable capacitance?

### 37-10 SAMPLE DATA

### DATA: CAPACITORS IN CIRCUITS

Resistance: 99.5 kΩ Capacitance: 100 μF Time Constant $\tau$ = RC = 9.95 s

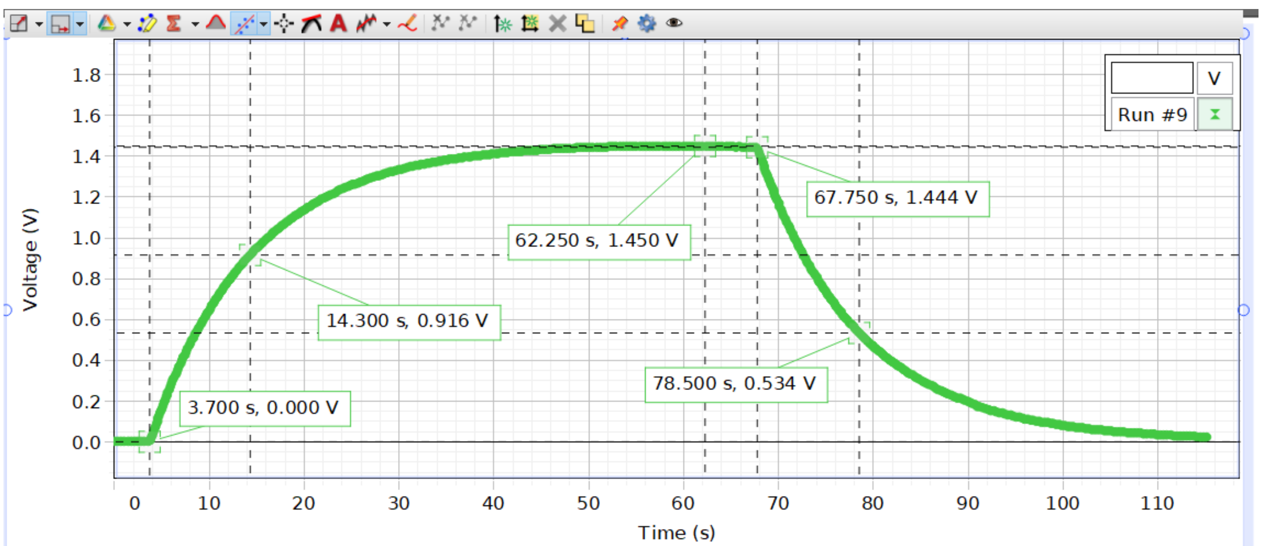
Battery potential: 1.620 V

***Table 1: Time for charging and discharging a capacitor (see figure 10)***

| Trial No. | **Vmax** **V** | **$V_1$ =0.632* Vmax** **V** | **$V_2$ = 0.368 Vmax** **V** | Measured Charging time $t_C$ *0 – $V_1$* **s** | Measured Discharging time $t_D$ *Vmax – $V_2$* **s** | Percent error in time of charging | Percent error in time of discharging |
|---|---|---|---|---|---|---|---|
| 1 | 1.450 | 0.9164 | 0.5336 | 14.30-3.70 = 10.6 | 78.5-67.75 =10.75 | 6.5 | 8.0 |
| | | | | | | | |

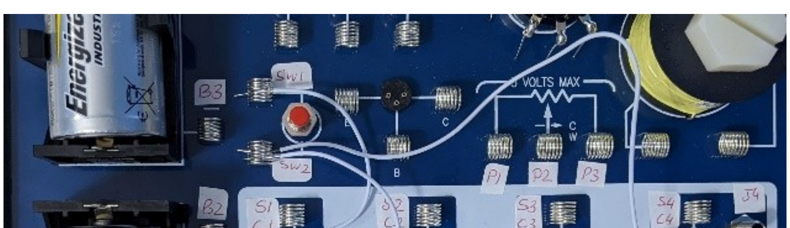
100*(10.6-9.95) / 9.95 = 6.5

100*(10.75 – 9.95) / 9.95 = 8.0




RC charging and discharging

Figure 10: Single Capacitor



**Two Capacitors in Series and Parallel:**

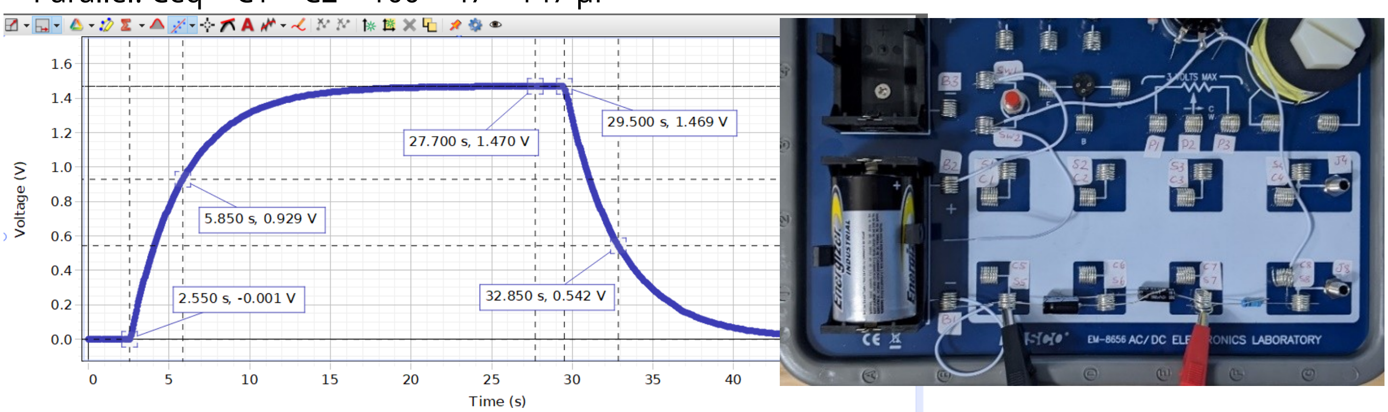
Resistance: 99.5 kΩ　　　Capacitor 1: 100 μF　　　Capacitor 2: 47 μF

***Table 2: Time for charging and discharging two capacitors in series and parallel.***

| Type of Circuit Cap.s in: | Equiv. Cap. **Ceq** **μF** | Calc. time for $t_C$ and $t_D$ =*RCeq* **s** | **Vmax** **V** | **$V_1$ =0.632* Vmax** **V** | **$V_2$ = 0.368 Vmax V** | Meas. Charging time $t_C$ **s** | Meas. Dis-charging time $t_D$ **s** | Percent error in time of charging | Percent error in time of dis-charging |
|---|---|---|---|---|---|---|---|---|---|
| Series Fig. 11 | 31.97 | 3.2 s | 1.470 | 0.929 | 0.541 | 5.850-2.550= 3.30 | 32.850-29.500= 3.35 | 3.0 | 4.7 |
| Parallel Fig. 12 | 147 | 14.7 s | 1.272 | 0.804 | 0.468 | 15.750-3.400= 12.35 | 72.550-57.100= 15.45 | 16.3 | 5.1 |

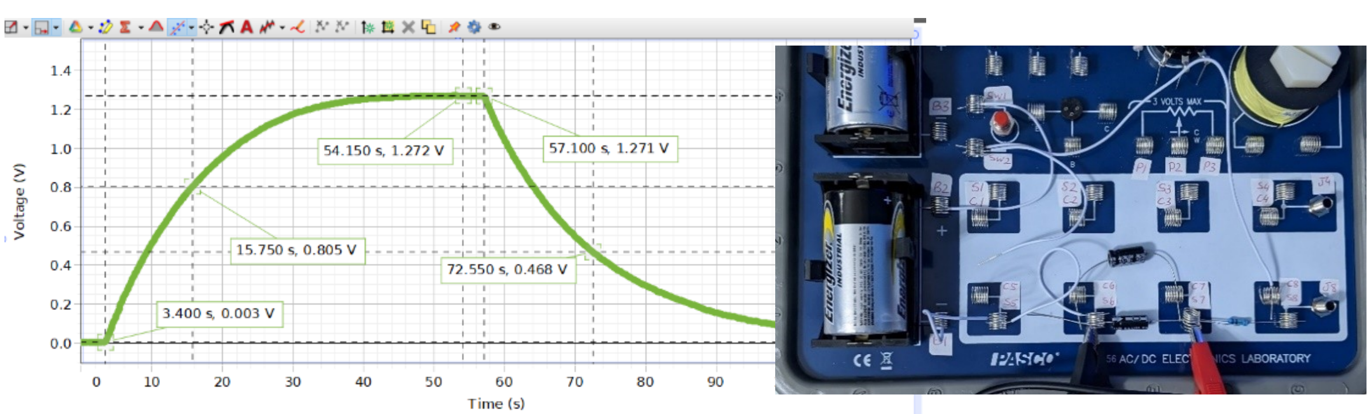
Series: 1/Ceq = 1/C1 + 1/C2 = 1/100 + 1/47 = 0.031277　　Ceq = 1/ 0.031277 = 31.97 μF

Parallel: Ceq = C1 + C2 = 100 + 47 = 147 μF




RC series charging and discharging

Figure 11: Two Capacitors in Series




RC parallel charging and discharging

Figure 12: Two Capacitors in Parallel

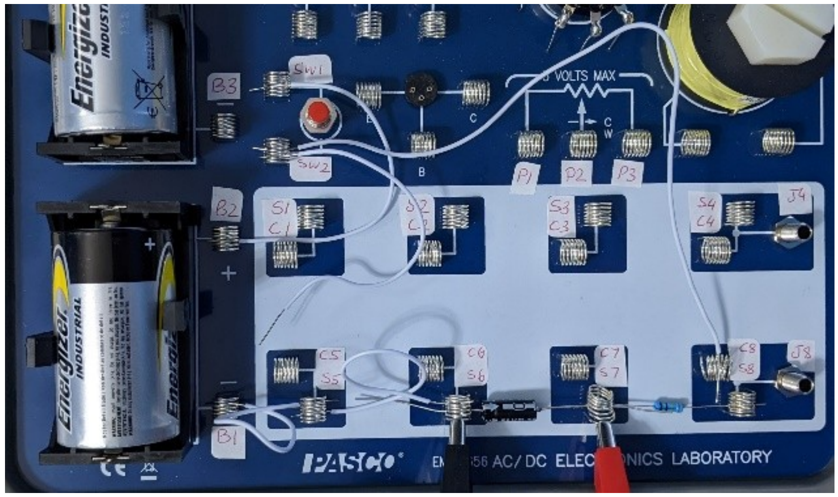



Figure 13:

Single Capacitor

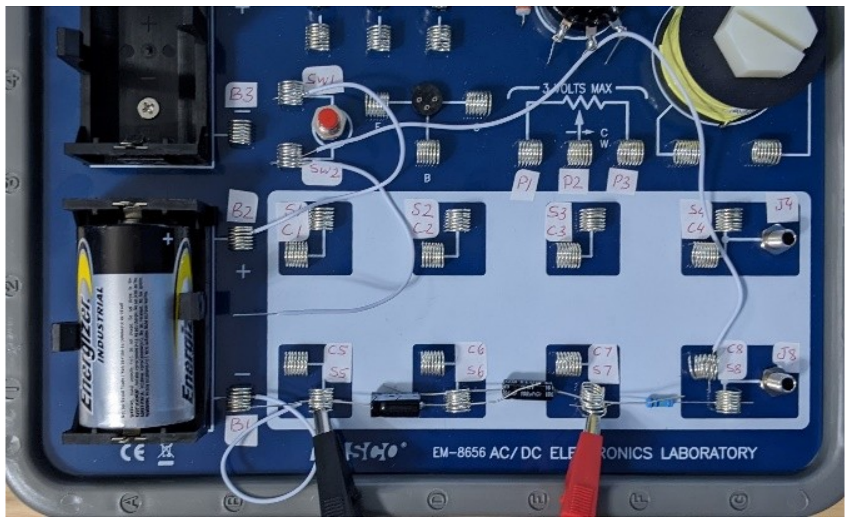



Figure 13B:

Two Capacitors in Series


Figure 13C:

Two capacitors in parallel